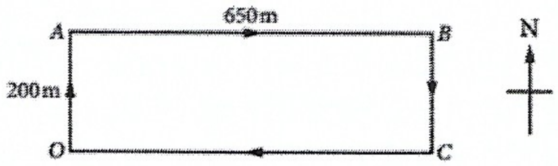
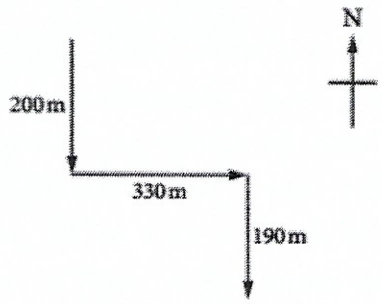
Motion and Forces

1. Jaide is going for a walk around the block. She starts at O and moves to point A and then to point B, where she stops to have a rest.



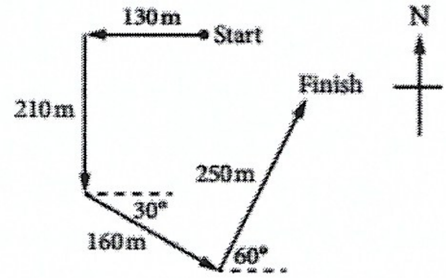
   a. What is the distance Jaide has travelled from O to B?
   b. What is Jaide's displacement from O to B, correct to one decimal place, and what is her direction? Jaide has taken 12 minutes to reach point B.
   c. What is Jaide's average speed in metres per second, correct to one decimal place?
   d. What is Jaide's average velocity in metres per second, correct to one decimal place, and what is her direction?

2. Courtney walks $200\ m$ south, then $330\ m$ east, and finally $190\ m$ south (diagram shown is not to scale). What is Courtney's displacement for her entire walk, correct to one decimal place, and what is the direction?



3. A mountain-climbing expedition established its base camp and two intermediate camps at positions A and B. Camp A is $8400\ m$ west of and $1800\ m$ above base camp. Camp B is $5900\ m$ west of and $850\ m$ higher than Camp A. What is the displacement of camp B from the base camp and what is its angle of elevation? Give answers correct to one decimal place.

4. Kenneth is talking a dog for a walk along the path in the local park, as shown in the diagram at the bottom (not to scale). What is the magnitude and direction of Kenneth's resultant displacement, correct to the nearest whole number?



5. While exploring a recently discovered cave system, a spelunker (cave explorer) starts at the entrance and makes the following movements: $85\ m$ north, $190\ m$ east, $250\ m\ N45°E$, and $100\ m$ south. What is the spelunker's final displacement from the cave entrance and what is their direction?

6. Brianna can swim in still water at a rate of $6.0\ km\ h^{-1}$. If she swims in a river flowing at $4.5\ km\ h^{-1}$ and keeps her direction (with respect to the water) perpendicular to the flow, then what is the magnitude of her velocity (correct to one decimal place) with respect to the riverbank?

7. a. A river flows at $5\ km\ h^{-1}$ and Rhani rows at $10\ km\ h^{-1}$. In what direction should Rhani row to go straight across the river?
   b. If the river flows at $10\ km\ h^{-1}$ and Rhani rows at $5\ km\ h^{-1}$, then in what direction should she row to go straight across the river?

8. Youlin and Nick are riding in a boat that has a speed relative to the water of $3.00\ m\ s^{-1}$. The boat points at an angle of 35° to the shore, moving upstream on a river that is flowing at $0.75\ m\ s^{-1}$. How much time does it take for the boat to reach the opposite shore, correct to two decimal places, if the river is $50\ m$ wide?

9. a. Raphacla is watching an aircraft that is flying at a velocity of $100\ m\ s^{-1}\ N$ with respect to the flow of air. If the velocity of the wind is $10\ m\ s^{-1}$ north, what is the resultant velocity of the aircraft relative to Raphaela?

b. Later, she observes a second aircraft flying with a velocity of $125\ m\ s^{-1}\ N$ with respect to the flow of the air. If the flow of the air has a velocity of $10\ m\ s^{-1}$ south, then the resultant velocity of this aircraft is $115\ ms^{-1}$. What is the direction of this resultant velocity, relative to Raphaela?

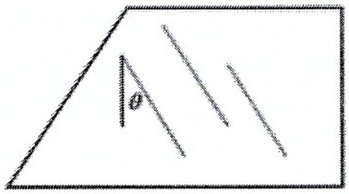
c. The next day, Raphaela observes another aircraft flying with a velocity of $100\ m\ s^{-1}N$ which encounters a wind coming from the side at a rate of $25\ m\ s^{-1}W$. What is the resultant velocity of this aircraft relative to Raphaela, correct to one decimal place, and what is its direction?

10. Let $\vec{\imath}$ and $\vec{\jmath}$ be unit vectors in the directions of the east and north respectively. Mitchell is swimming in the ocean and his velocity $\overrightarrow{v_1}\ m\ s^{-1}$ is given by the vector $\overrightarrow{v_1} = 1.0\vec{\imath} - 2.0\vec{\jmath}$. The ocean's current has a velocity $\overrightarrow{v_2}\ m\ s^{-1}$ where $\overrightarrow{v_2} = -0.5\,\vec{\imath} + 2.5\vec{\jmath}$. What is the magnitude and direction of Mitchell's resultant velocity $\vec{v}\ m\ s^{-1}$, correct to one decimal place, and what is the direction?

11. Let $\vec{\imath}$ and $\vec{\jmath}$ be unit vectors in the directions of east and north respectively. Given answers correct to two decimal places where required.
    a. Express each vector in the form $x\vec{\imath} + y\vec{\jmath}$.
        i. $\overline{OA} = 5.0\ m$ at $053°T$
        ii. $\overline{AB} = 6.0\ m$ at $315°T$
        iii. $\overline{BC} = 4.0\ m$ at $240°T$
        iv. $\overline{CD} = 3.0\ m$ at $150°T$
    b. Express the sum of the four displacement vector in the form $x\vec{\imath} + y\vec{\jmath}$, with values correct to two decimal places.
    c. What is the magnitude and direction of the resultant vector $\overline{OD}$, correct to two decimal places, and what is its direction?

12. Let $\vec{\imath}$ and $\vec{\jmath}$ be unit vectors in the directions of east and north respectively. Morgan and Eilish finish their last day of school and immediately decide to take a road trip across the desert in Eilish's new car. Their trip involves the following movements: $43\ km$ at $337°T$, $65\ km$ at $270°T$ and $22\ km$ at $93°T$. Eilish's car breaks down after the last leg of the trip. At this time, how far are Morgan and Eilish from the school and in what direction?

13. Johanna is driving along the freeway at $108\ km\ h^{-1}$ when it begins to rain. She observes the raindrops running down the driver's side window. Calculate the angle that the raindrops make with the vertical window, as seen by Johanna, if the raindrops have a speed of $10\ m\ s^{-1}$ relative to the Earth's surface. Assume that the raindrops fall vertically down, relative to the Earth's surface.



14. Bus #1 is moving at a speed of $50\ km\ h^{-1}$, while Bus #2 is moving at $30\ km\ h^{-1}$ in the opposite direction. What is the relative velocity of Bus #1 with respect to Bus #2?

15. On her way home from school, Teluila walks along three streets after exiting the school gate. She walks $240\ m$ east, $720\ m$ north and $75\ m$ east. What is the magnitude of Teluila's resultant displacement?

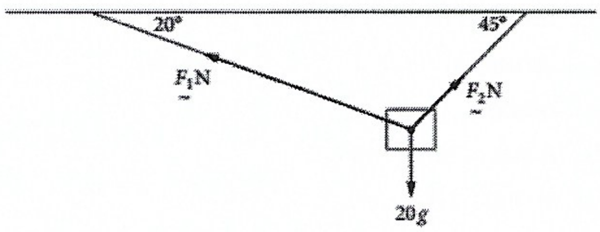
16. The pilot of an aircraft flying due south is notified by the flight controller that there is a second aircraft flying due north in the same general area and at the same altitude. The pilot is told that the northbound aircraft is currently located at a position that is $12.6\ km$, $170°T$ with respect to the pilot's aircraft.
    a. How many kilometres to the south is the second aircraft?
    b. How many kilometres to the east is the second aircraft?
    c. If the two aircrafts both have an airspeed of $300\ km\ h^{-1}$, how much time (in seconds) will elapse before they are *side by side*?

17. An aircraft drops a package of emergency rations to a family stranded in the floods. The aircraft is travelling horizontally at $45.0\ m\ s^{-1}$ and is $100\ m$ above the ground. A parachute

allows the package to fall with constant speed and hit the ground 10 s after release.

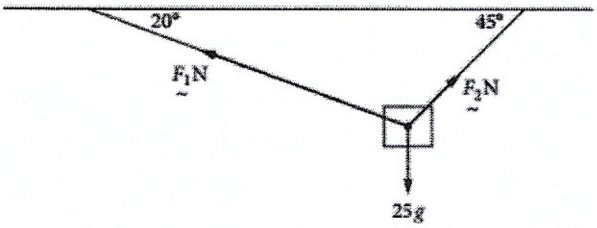
a. Find where the package hits the ground, relative to the point from where it was dropped, to the nearest metre.
b. Find the velocity of the package just before it hits the ground, correct to one decimal place.

18. Forces are acting on an object. Calculate the resultant force acting on the object for each of the following:
    a. $\overrightarrow{F_1} = 150N$ north, $\overrightarrow{F_2} = 120N$ south
    b. $\overrightarrow{F_1} = 67N$ east, $\overrightarrow{F_2} = 83N$ west
    c. $\overrightarrow{F_1} = 320N$ east, $\overrightarrow{F_2} = 210N$ west, $\overrightarrow{F_3} = 140N$ east
    d. $\overrightarrow{F_1} = 64N$ south, $\overrightarrow{F_2} = 56N$ north, $\overrightarrow{F_3} = 48N$ south.
19. Forces are acting on an object. Calculate the magnitude and direction of the resultant force acting on the object for each of the following:
    a. $\overrightarrow{F_1} = 160N$ north, $\overrightarrow{F_2} = 120N$ west
    b. $\overrightarrow{F_1} = 80N$ east, $\overrightarrow{F_2} = 90N$ south
    c. $\overrightarrow{F_1} = 125N$ west, $\overrightarrow{F_2} = 85N$ south
    d. $\overrightarrow{F_1} = 340N$ north, $\overrightarrow{F_2} = 110N$ east
20. Two tugboats are applying simultaneous forces to a container ship as it attempts into port. One tugboat is applying a force of $3200N$ in an easterly direction and the second tugboat is applying a $2300N$ force directly south. What is the magnitude and direction of the resultant force being applied to the container ship?
21. Two forces are acting on an object: calculate the resultant force $\vec{F}$ acting on the object for each of the following. Give answers correct to two decimal places.
    a. $\overrightarrow{F_1} = 100N$ acting at $N40°E$ and $\overrightarrow{F_2} = 90N$ acting at $N35°W$
    b. $\overrightarrow{F_1} = 200N$ acting at $N39°W$ and $\overrightarrow{F_2} = 160N$ acting at $S40°W$
    c. $\overrightarrow{F_1} = 15N$ acting at $152°T$ and $\overrightarrow{F_2} = 23N$ acting at $065°T$
    d. $\overrightarrow{F_1} = 2050N$ acting at $037°T$ and $\overrightarrow{F_2} = 1560N$ acting at $226°T$
22. Three forces $\overrightarrow{F_1}, \overrightarrow{F_2}$ and $\overrightarrow{F_3}$ are acting on an object: $\overrightarrow{F_1} = 200N$ at $057°T$, $\overrightarrow{F_2} = 220N$ at $170°T$ and $\overrightarrow{F_3} = 150N$ at $245°T$.
    a. Resolve each force into horizontal $\vec{\imath}$ and vertical $\vec{\jmath}$ components.
    b. Determine the resultant force $\vec{F}$. Give answers in component form.
23. Three forces are acting on an object: $\overrightarrow{F_1} = 250N$ at $N45°E$, $\overrightarrow{F_2} = 270N$ at $S35°W$ and $\overrightarrow{F_3} = 350N$ at $S25°E$.
    a. Resolve each force into horizontal $\vec{\imath}$ components and vertical $\vec{\jmath}$ components.
    b. Determine the resultant force $\vec{F}$. Give answers in component form.
24. A particle of mass $20kg$ is suspended by two strings attached to two points in the same horizontal plane. If the two strings make angles of 20° and 45° respectively to the horizontal, find the magnitude of the tension force in each string, in newtons correct to two decimal places.



26. a. A particle of mass $25kg$ is suspended by two strings attached to two points in the same horizontal plane. If the two string s make angles of 27° and 42° respectively to the horizontal, draw a force diagram and find the magnitude of the tension forces, F_1 and F_2 respectively in each string, in newtons correct to two decimal places.
    b. A particle of mass $56kg$ is suspended by two string attached to two strings attached to two points in the same horizontal plane. If the two strings make angles of 18° and 54° respectively to the horizontal, draw a force diagram and find the magnitude of the tension forces, F_1 and F_2 respectively in each string, in newtons correct to two decimal places.

26. A particle of mass $25kg$ is suspended by two strings attached to two points in the same horizontal plane. If the two strings make angles of $20°$ and $45°$ respectively to the horizontal, use component form to find the magnitude of the tension forces, F_1 and F_2 respectively in each string, in newtons correct to one decimal place.

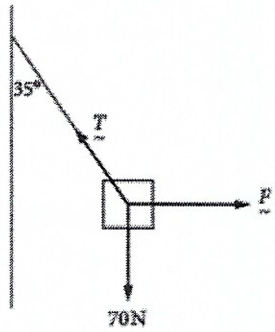


28. a. A particle mass $15kg$ is suspended by two strings attached to two points in the same horizontal plane. If the two strings make angles of $22°$ and $38°$ respectively to the horizontal, draw a force diagram and use component form to find the magnitude of the tension forces, F_1 and F_2 respectively in each string, in newtons correct to two decimal places.

b. A particle of mass $45kg$ is suspended by two strings attached to two points in the same horizontal plane. If the two strings make angles of $36°$ and $34°$ respectively to the horizontal, draw a force diagram and use component form to find the magnitude of the tension forces, F_1 and F_2 respectively in each string, in newtons correct to two decimal places.

28. An object that is being pulled vertically downwards by a force of $70N$ is attached to a wall by string of negligible mass. The object is also being pulled to the right by a horizontal force so that object is not moving and the string makes an angle of $35°$ with the vertical, as shown.

Find the magnitudes of the horizontal force $\vec{F}N$ and the tension force in the string, $\vec{T}N$. Given answers correct to two decimal places.



29. An object that is being pulled vertically downwards by a force of $250N$ is attached to a wall by string of negligible mass. The object is also being pulled to the right by a horizontal force so that the object is not moving and the string makes an angle of $22°$ with the vertical.

   a. Complete the force diagram for this situation. Label the horizontal force $\vec{F}N$ and the tension in the string $\vec{T}N$.
   b. Find the magnitudes of the horizontal force, $\vec{F}N$, and the tension in the string, $\vec{T}N$.

30. Three forces are acting on an object simultaneously in the same plane. The resultant force is $\vec{F} = 360N$ due north. Which of the following combinations of forces will give this resultant force?

   a. $\overrightarrow{F_1} = 440N$ south, $\overrightarrow{F_2} = 560N$ north, $\overrightarrow{F_3} = 480N$ south
   b. $\overrightarrow{F_1} = 120N$ north, $\overrightarrow{F_2} = 250N$ south, $\overrightarrow{F_3} = 480N$ north
   c. $\overrightarrow{F_1} = 440N$ north, $\overrightarrow{F_2} = 560N$ south, $\overrightarrow{F_3} = 480N$ north
   d. $\overrightarrow{F_1} = 120N$ south, $\overrightarrow{F_2} = 240N$ north, $\overrightarrow{F_3} = 480N$ south

32. a. Two tugboats are towing a large boat of mass $15000kg$ in an easterly direction towards a dock. The first tugboat pulls with a force of $\overrightarrow{F_1} = 8000N$ at an angle of $28°$ north of the forward motion while the second tugboat pulls with force of $\overrightarrow{F_2} = 8250N$ at an angle $\theta°$ south of the forward motion. Given that there is a resistive force of $4200N$ opposing the eastern motion of the large boat, calculate the total easterly force acting on the large boat, to the nearest newton.

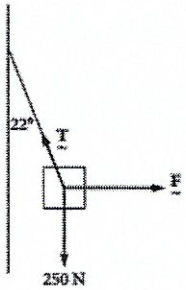
b. Two tugboats are towing a large boat of mass $M\ kg$ back to shore. The first tugboat pulls with a force of $\overrightarrow{F_1}N$ at an angle of $27°$ north of the forward of the forward motion and the second tugboat pulls with a force of $\overrightarrow{F_2}N$ at an angle $30°$ south of the forward motion. If the large boat is moving with constant velocity and there is a resistive force of $5500N$ opposing this motion, calculate the magnitudes of the two forces $\overrightarrow{F_1}$ and $\overrightarrow{F_2}$ to the nearest newton.

32. A section of a new bridge is being moved by four cranes that will move it horizontally into position. The chains that are connected from the cranes to the bridge section exert forces that are acting on the bridge simultaneously and in the same plane. The four forces are $\overrightarrow{F_1} = 2050N$ at $037°T$, $\overrightarrow{F_2} = 1560N$ at $130°T$, $\overrightarrow{F_3} = 1650N$ acting at $237°T$ and $\overrightarrow{F_4} = 1930N$ acting at $316°T$.
    a. Resolve each force into horizontal $\vec{i}$ and vertical $\vec{j}$ components.
    b. Calculate the magnitude and direction of the resultant force applied to the bridge section.

Answer:

1. **a.** $850\ m$; **b.** Jaide's displacement from O to B, correct to one decimal place, is $680.1\ m$ from O in a direction of $N72.9°E$; **c.** $1.2\ m\ s^{-1}$; **d.** Jaide's average velocity, in metres per second, correct to one decimal place, is $0.9\ m\ s^{-1}$ in a direction of $N72.9°E$
2. Courtney's displacement for her entire walk is $510.9\ m$ in the direction $S40.2°E$.
3. The displacement of camp B from the base camp is $14543.5\ m$ at an angle of elevation of $10.5°$.
4. The magnitude and direction of Kenneth's resultant displacement is $152\ m$ in the direction $S61°E$.
5. The spelunker's final displacement from the cave entrance is $401\ m$ in the direction $N66°E$.
6. If she swims in a river flowing at $4.5\ km\ h^{-1}$ and keeps her direction (with respect to the water) perpendicular to the current, the magnitude of her velocity with respect to the riverbank is $7.5\ km\ h^{-1}$.
7. **a.** Head at angle of $60°$ to the riverbank upstream; **b.** not possible
8. The time it takes for the boat to reach the opposite shore, if the river is $50\ m$ wide, is 29.06 seconds.
9. **a.** If the velocity of the wind is $10\ m\ s^{-1}$ north, the resultant velocity of the aircraft relative to Rapahela is $110\ m\ s^{-1}$ north; **b.** If the head wind has a velocity of $10\ m\ s^{-1}$ south, the resultant velocity of this aircraft relative to Rapahela is $115\ m\ s^{-1}$ in the direction north; **c.** The resultant velocity of this aircraft relative to Rapahela is $103.1\ m\ s^{-1}$ in the direction $N14.0°W$.
10. The magnitude and the direction of Mitchell's resultant velocity, $\vec{v}\ m\ s^{-1}$ is $\vec{v} = 0.7\ m\ s^{-1}$ in the direction 045 or $45°T$.
11. **a.** i. $\overline{OA} = 3.99\vec{i} + 3.01\vec{j}$, ii. $\overline{AB} = -4.24\vec{i} + 4.24\vec{j}$, iii. $\overline{BC} = -3.46\vec{i} - 2\vec{j}$, iv. $\overline{CD} = 1.5\vec{i} - 2.06\vec{j}$; **b.** $\overline{OD} = -2.11\vec{i} + 3.19\vec{j}$; **c.** The magnitude and direction of the resultant vector $\overline{OD}$ is $3.88\ m$ at $145.29°T$.
12. At this time, Morgan and Eilish are $71.1\ km$ from the school and in the direction $302.7°T$.
13. 72°
14. $80\ km\ h^{-1}$ in the direction of bus 1.
15. 786 m
16. **a.** The second aircraft is $12.4\ km$ to the south; **b.** The second aircraft is $2.2\ km$ to the east; **c.** $74.5\ s$ will have elapsed before the aircrafts are side by side.
17. **a.** The package hits the round $450.0\ m$ relative to the point from where it was dropped; **b.** The velocity is $46.1\ m\ s^{-1}$ in a direction of $77.5°$ to the vertical.
18. **a.** $\overrightarrow{F_1} + \overrightarrow{F_2} = 30N$, north; **b.** $\overrightarrow{F_1} + \overrightarrow{F_2} = -16N$, west; **c.** $\overrightarrow{F_1} + \overrightarrow{F_2} + \overrightarrow{F_3} = 250N$, east; **d.** $\overrightarrow{F_1} + \overrightarrow{F_2} + \overrightarrow{F_3} = 56N$, south
19. **a.** $200.00N$ at $N36.87°W$; **b.** $120.42N$ at $S41.63°E$; **c.** $151.16N$ at $S55.78°W$; **d.** $357.35N$ at $N17.93°E$.
20. The resultant force being applied to the container ship is $3941N$ in the direction $S54°E$.
21. **a.** $\vec{F} = 150.86N$, acting in the direction $N4.81°E$; **b.** $\vec{F} = 231.06N$, acting in the direction $N81.82°W$; **c.** $\vec{F} = 28.11N$, acting in the direction $97.2°T$; **d.** $\vec{F} = 564.66N$, acting in the direction $011.39°T$
22. **a.** $\overrightarrow{F_1} = 167.73\vec{i} + 108.93\vec{j}$, $\overrightarrow{F_2} = 38.20\vec{i} - 216.66\vec{j}$, $\overrightarrow{F_3} = -135.95\vec{i} - 63.39\vec{j}$; **b.** $\vec{F} = 69.99\vec{i} - 171.12\vec{j}$
23. **a.** $\overrightarrow{F_1} = 176.78\vec{i} + 176.78\vec{j}$, $\overrightarrow{F_2} = -154.87\vec{i} - 221.17\vec{j}$, $\overrightarrow{F_3} = 147.92\vec{i} - 317.21\vec{j}$
24. $|\overrightarrow{F_1}| = 152.92N$, $|\overrightarrow{F_2}| = 203.22N$

25. **a.** $|\vec{F_1}| = 195.02N$, $|\vec{F_2}| = 233.83N$; **b.** $|\vec{F_1}| = 339.21N$, $|\vec{F_2}| = 548.81N$
26. $|\vec{F_1}| = 191.2N$, $|\vec{F_2}| = 254.0N$
27. **a.** $|\vec{F_1}| = 133.76N$, $|\vec{F_2}| = 157.38N$; **b.** $|\vec{F_1}| = 389.06N$, $|\vec{F_2}| = 379.68N$
28. $|\vec{T}| = 85.45N$; $|\vec{F}| = 49.01N$
29. **a.** see below; **b.** $|\vec{T}| = 269.63N$; $|\vec{F}| = 101.01N$.



30. **C.**
31. **a.** $\vec{F} = 10209N$; **b.** $|\vec{F_1}| = 3279$, $|\vec{F_2}| = 2977$
32. **a.** $\vec{F_1} = 1233.72\vec{\imath} + 1637.20\vec{\jmath}$, $\vec{F_2} = 1195.03\vec{\imath} - 1002.75\vec{\jmath}$, $\vec{F_3} = -1383.81\vec{\imath} - 898.65\vec{\jmath}$, $\vec{F_4} = -1340.69\vec{\imath} + 1388.33\vec{\jmath}$; **b.** $\vec{F} = 1162N$ at $345°T$ on a bearing is $345.26°T$.